

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped a production self-service analytics agent over 67 BigQuery tables — multi-agent architecture with Context Researcher → SQL Drafter → Validator → Devil's Advocate sub-agents and context-isolated validation, **governed by 8 deterministic guardrail rules and a 23-case golden-query evaluation suite**. Approved by Sr. Director and product leadership and currently the only AI skill in active use by business teams on Data Ventures.
- Built an autonomous Jira ticket-resolution agent: **6-gate triage**, MCP-driven context gathering, FAISS+BM25 retrieval over 11,000 historical SQL queries, **self-healing execution with a 10-retry budget**. Closed 400+ data requests across 8 categories over 15 months — cut turnaround from 30–60 minutes per request (and up to a full day for complex ones) to under 10 minutes each. Stakeholders never knew an agent was answering them.
- Built a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; **state persists to BigQuery so the agent has no LLM-memory reliance**. Designed a v2 architecture as a 3-tier event-driven platform (reactive events, threshold KPIs, PM-owned themes) self-configuring via a BigQuery event registry. New event types ship without code deploys.
- Built a hybrid orchestrator on top of the analytics agent. Routes business questions through two backends in parallel: a CubeJS semantic-layer skill (co-built with the DS and product team) and the context-engineered agent itself. **Compares outputs, reasons about disagreements, and feeds discrepancies back to refine the semantic layer's definitions over time**.
- Replaced ~20 distinct recruitment workflows I'd been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); cut data-request turnaround from 1–24 hours to 5–10 minutes; Making a Difference Award.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven **multi-LLM (OpenAI + Gemini)** chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

Capstone engagement with FTI Consulting, a global financial & business advisory firm

- Built a RAG-to-proposal generator that automated drafting of client engagement proposals for FTI's advisory teams; deployed on AWS EC2, cut proposal turnaround ~30%; Best in Show — UChicago MSADS Capstone.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Rapidly prototyped and live-deployed a text-to-SQL copilot (Streamlit + Gemini 2.5 + RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Evaluation & reliability: golden-query evaluation suites, deterministic guardrail rules, offline testing of agent flows, output consistency checks, self-healing execution, decision logging & monitoring, A/B testing across LLM providers

Agentic AI: multi-agent architecture (planner / executor / reviewer / supervisor patterns), sub-agent orchestration, MCP (Model Context Protocol) server design, tool- & function-calling agents, memory-enabled agents, context & prompt engineering, event-driven agent workflows, headless cron agents

RAG & retrieval: FAISS + BM25 + RRF, ChromaDB semantic memory, multi-model routing across providers

LLMs & frameworks: Claude (Opus / Sonnet / Haiku — daily via Wibey, Walmart's internal Claude Code fork), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

Engineering & delivery: Python, FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores

Data & viz: BigQuery, dbt, SQL, PySpark, Airflow, Databricks, MySQL; Power BI (DAX, XMLA TMSL), Tableau, LookML, Streamlit, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL